

Building your own methods

INTERMEDIATE JAVA



Jim White
Java Developer

Combine everything!

- Create your own methods
 - Including parameters
- Control flow

Strings , arrays, ...



Check if number is even

```
class IsItEven {  
    public static void main(String[] args) {  
        int number = 2345;  
        // Call the method and print the result  
        System.out.println("Number " + number + " is even: " + isEven(number));  
    }  
  
    static boolean isEven(int n) {  
        return (n % 2) == 0; // Use modulo operator, that returns the remainder  
    }  
}
```

Number 2345 is even: false

Iterate through given String and print each letter

```
class PrintItOut {  
    public static void main(String[] args) {  
        String name = "Hi";  
        printLetters(name); // Call the method on name  
    }  
    static void printLetters(String str) {  
        for (int i = 0; i < str.length(); i++) { // Iterates through given String  
            System.out.println(str.charAt(i)); // Prints each letter out  
        }  
    }  
}
```

```
H  
i
```

Let's practice!

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Looping and methods

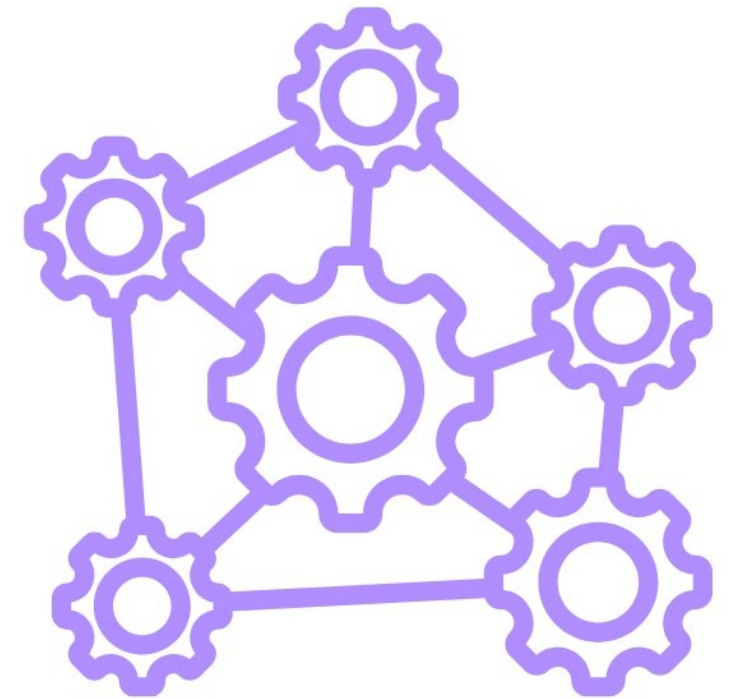
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Combine loops and custom methods

- Doubling values in an array
- Finding the longest word in a list
- ...



Loop through an array and square each element

```
class SquareIt {  
    public static void main(String[] args) {  
        int[] arr = {2, 3, 4, 5};  
        int[] squaredArr = squareArray(arr); // squaredArr contains 4, 9, 16, and 25  
    }  
  
    static int[] squareArray(int[] arr) {  
        for (int i = 0; i < arr.length; i++) { // Iterate through the array  
            arr[i] = arr[i] * arr[i]; // Square each element  
        }  
        return arr;  
    }  
}
```


Find the longest word in an array

```
class LongestWord {
    public static void main(String[] args) {
        String[] arr = {"Jo", "Mia", "Emma", "Tom"};
        // Call the method to find longest word
        String longest = longestWord(arr); // Value is Emma
    }
    public static String longestWord(String[] arr) {
        String longest = "";
        for (String word : arr) { // Iterate through the array
            if (word.length() > longest.length()) { // Check if word is longer than longest
                longest = word; // If word is longer, update longest
            }
        }
        return longest;
    }
}
```

Let's practice!
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Building a workflow

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Double odd numbers, halve even numbers

```
class DoubleOrHalve {
    public static void main(String[] args) {
        int[] numbers = {23, 78, 2263, 552};
        int[] newNumbers = doubleOddHalveEven(numbers); // Value is 46, 39, 4526, and 276
    }
    static int[] doubleOddHalveEven(int[] arr) {
        for (int i = 0; i < arr.length; i++) { // Loop through the array
            if (arr[i] % 2 == 0) {
                arr[i] /= 2; // Halve even numbers
            } else {
                arr[i] *= 2; // Double odd numbers
            }
        }
        return arr;
    }
}
```

Method chaining

- Call methods directly after one another, on the same value
 - E.g., `variable.method1().method2()`

```
class CheckStrings {  
    public static void main(String[] args) {  
        String message = "Hello World!";  
        String input = "hello world!";  
  
        // Convert message to lower case, then compare with input  
        System.out.println(message.toLowerCase().equals(input));  
    }  
}
```

true

Structure

- Break code into clear steps
 - Use meaningful names
 - Keep logic together
- ➔ Makes code easier to understand, update, and maintain



Let's practice!

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Wrap-up

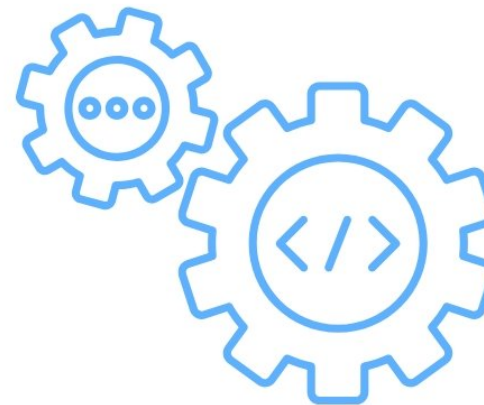
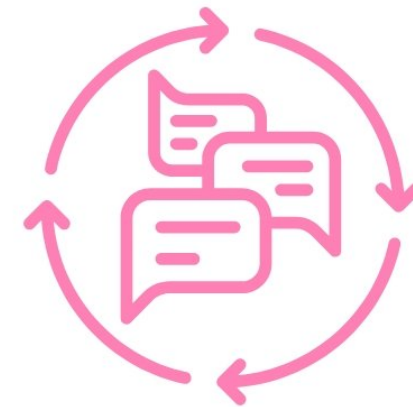
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What you've learned

- Custom methods including parameters
- Control flow
 - if - else if - else
 - Logic operators
 - Loops
 - for loop
 - for each loop
 - while loop
- Putting everything together



Where to next?

- Internet Sources
 - [Baeldung](#): website dedicated to Java topics and assistance
 - [Oracle's Java site](#): the stewards of Java technology (Oracle) web site
 - [dev.java](#): Java developer news, events, and other information
 - [Open JDK](#): community dedicated to a free and open-source implementation of Java
- Books
 - *Head First Java* by Sierra et al
 - *Learn Java In One Day* by Jamie Chan
 - *Java for Beginners* by Brady Ellison
- Additional online courses from **DataCamp**
 - Next one after this: [Introduction to Object-Oriented Programming in Java](#)

Congratulations!

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